

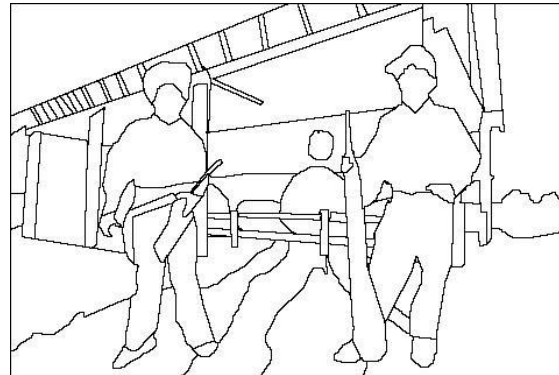
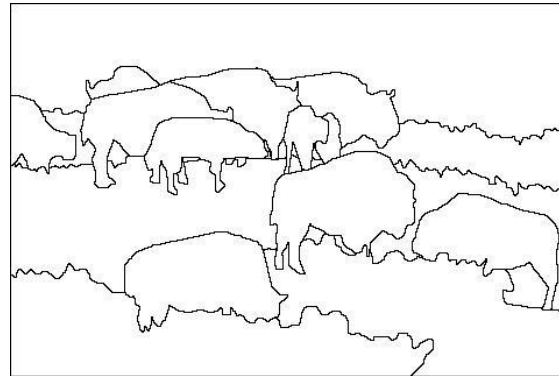
Digital Image Processing

Cluster Analysis

The goals of segmentation

- Separate image into coherent “objects”

image human segmentation



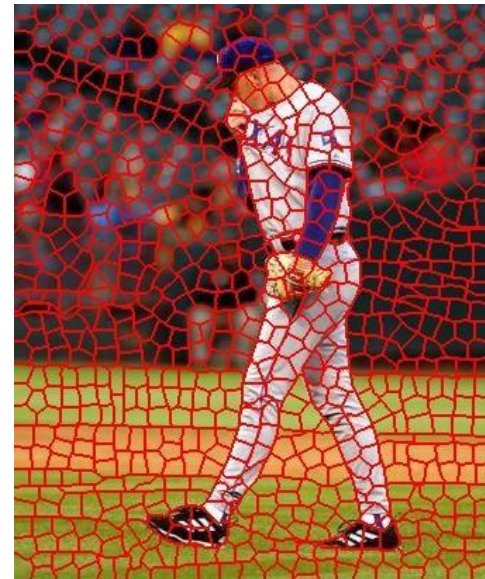
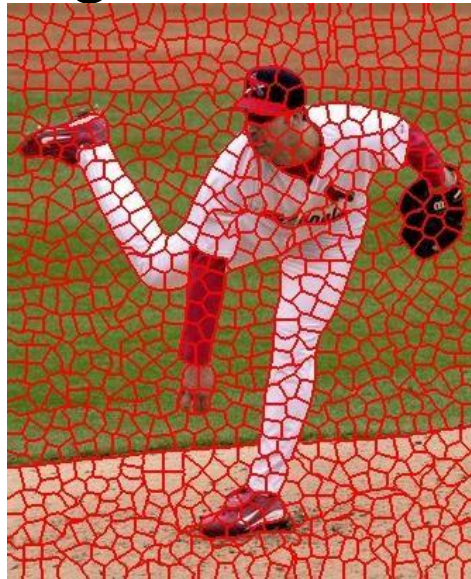
Berkeley segmentation database:

<http://www.eecs.berkeley.edu/Research/Projects/CS/vision/grouping/segbench/>

The goals of segmentation

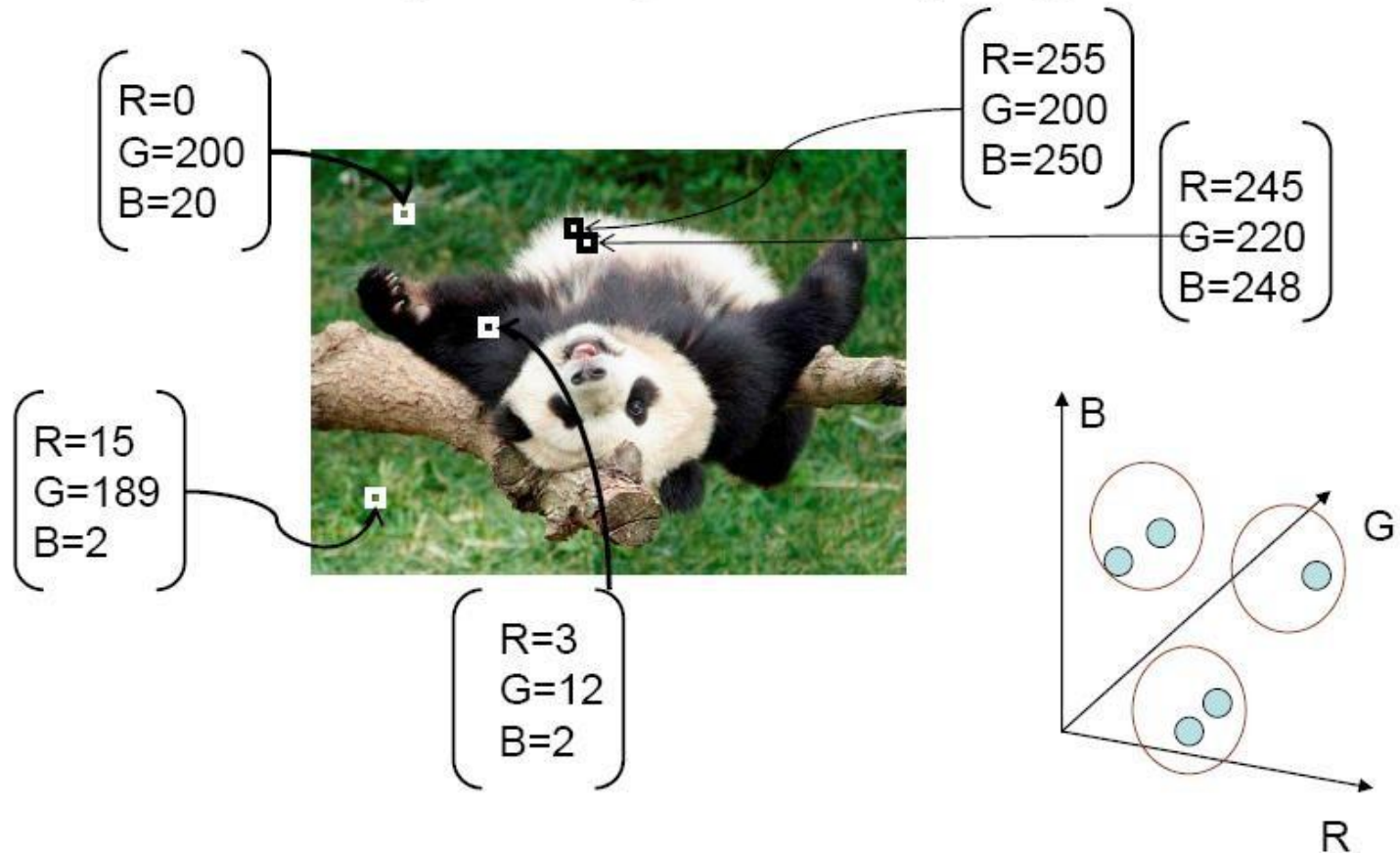
- Separate image into coherent “objects”
 - Top-down or bottom-up process?
 - Supervised or unsupervised?
- Group together similar-looking pixels for efficiency of further processing

“superpixels”



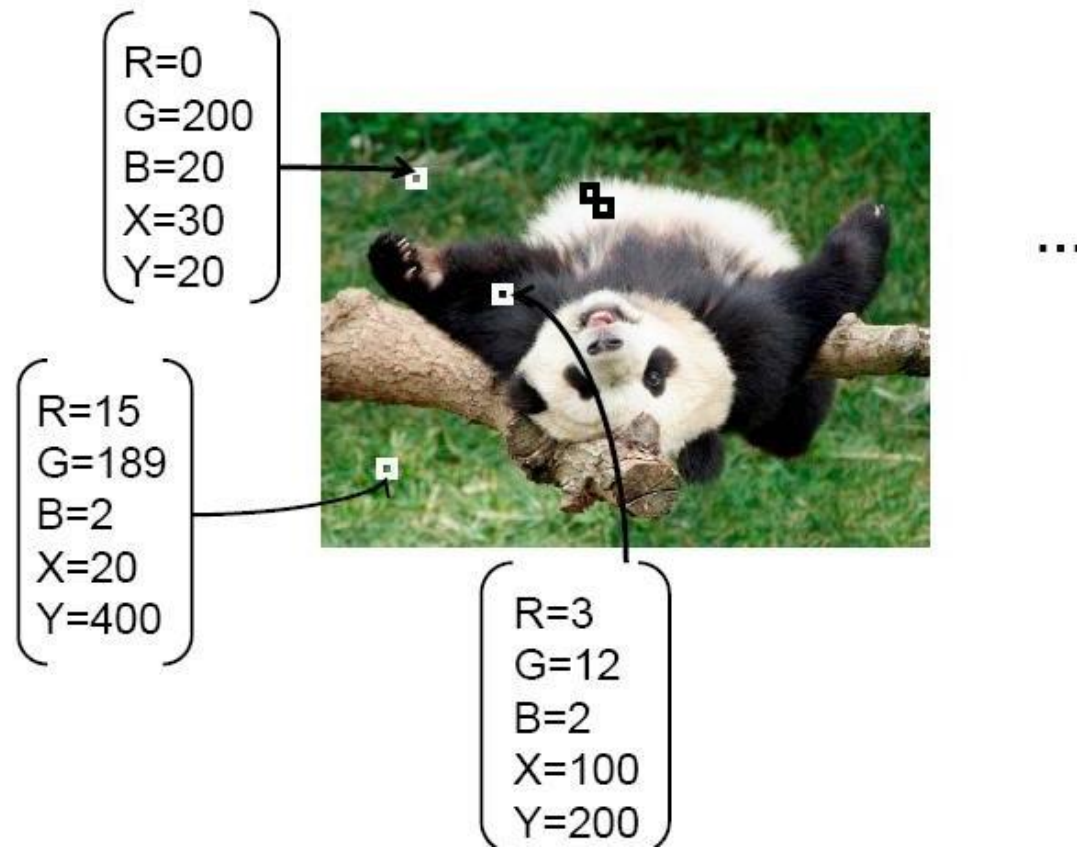
Segmentation as clustering

- Cluster similar pixels (features) together



Segmentation as clustering

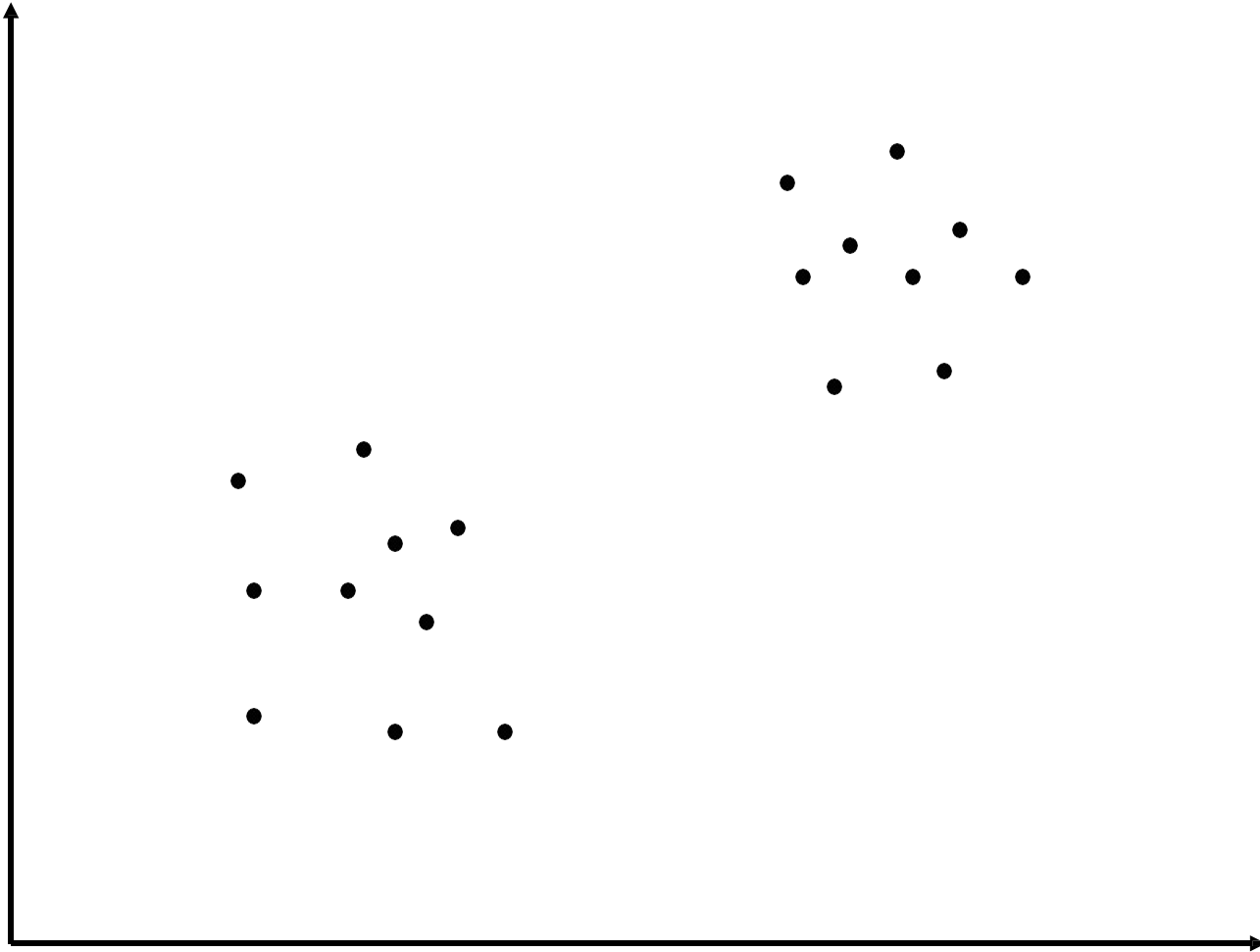
- Cluster similar pixels (features) together



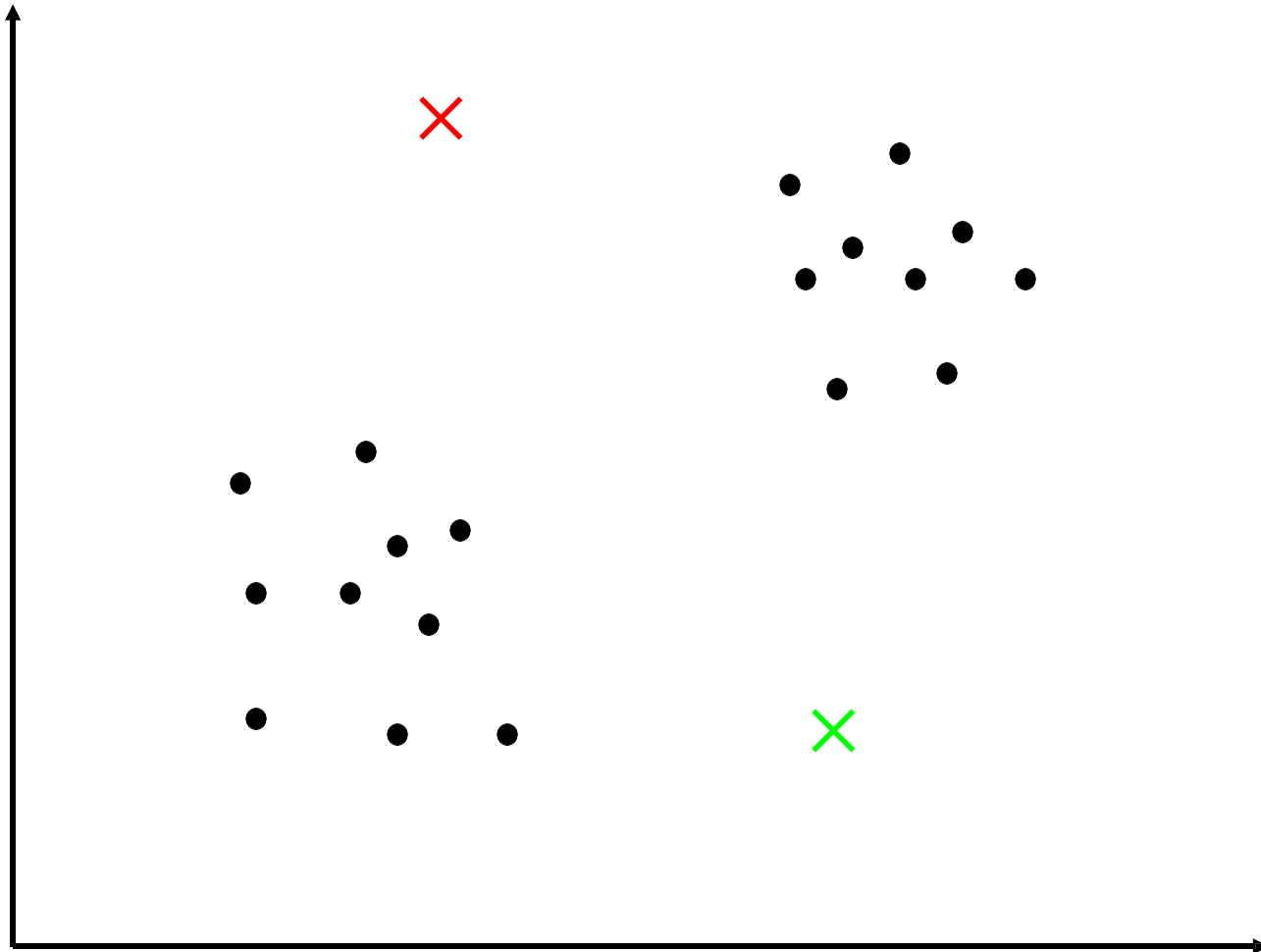
K-Means Clustering

1. Chose the number (K) of clusters and randomly select the centroids of each cluster.
2. For each data point:
 - Calculate the distance from the data point to each cluster.
 - Assign the data point to the closest cluster.
3. Recompute the centroid of each cluster.
4. Repeat steps 2 and 3 until there is no further change in the assignment of data points (or in the centroids).

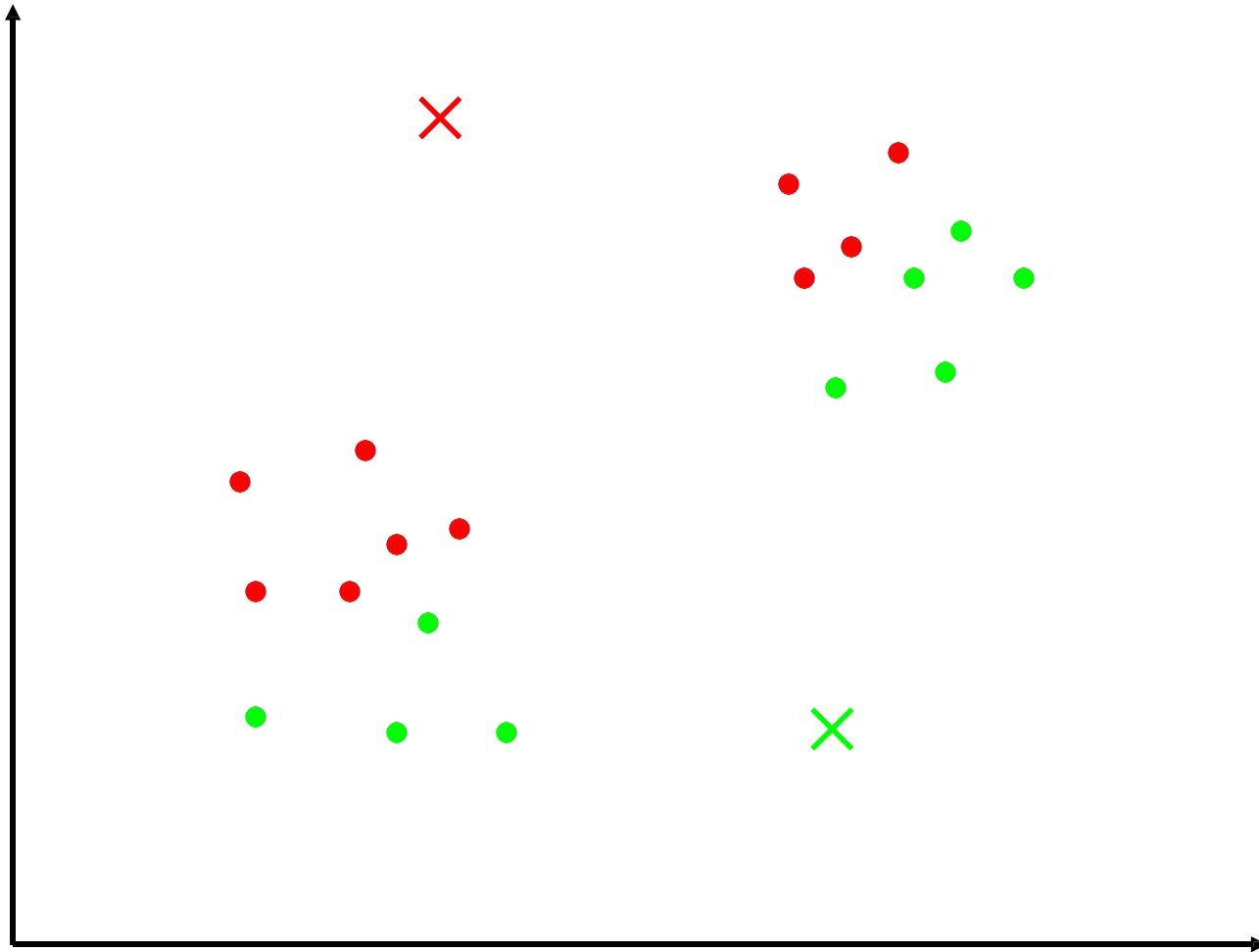
K-Means Clustering



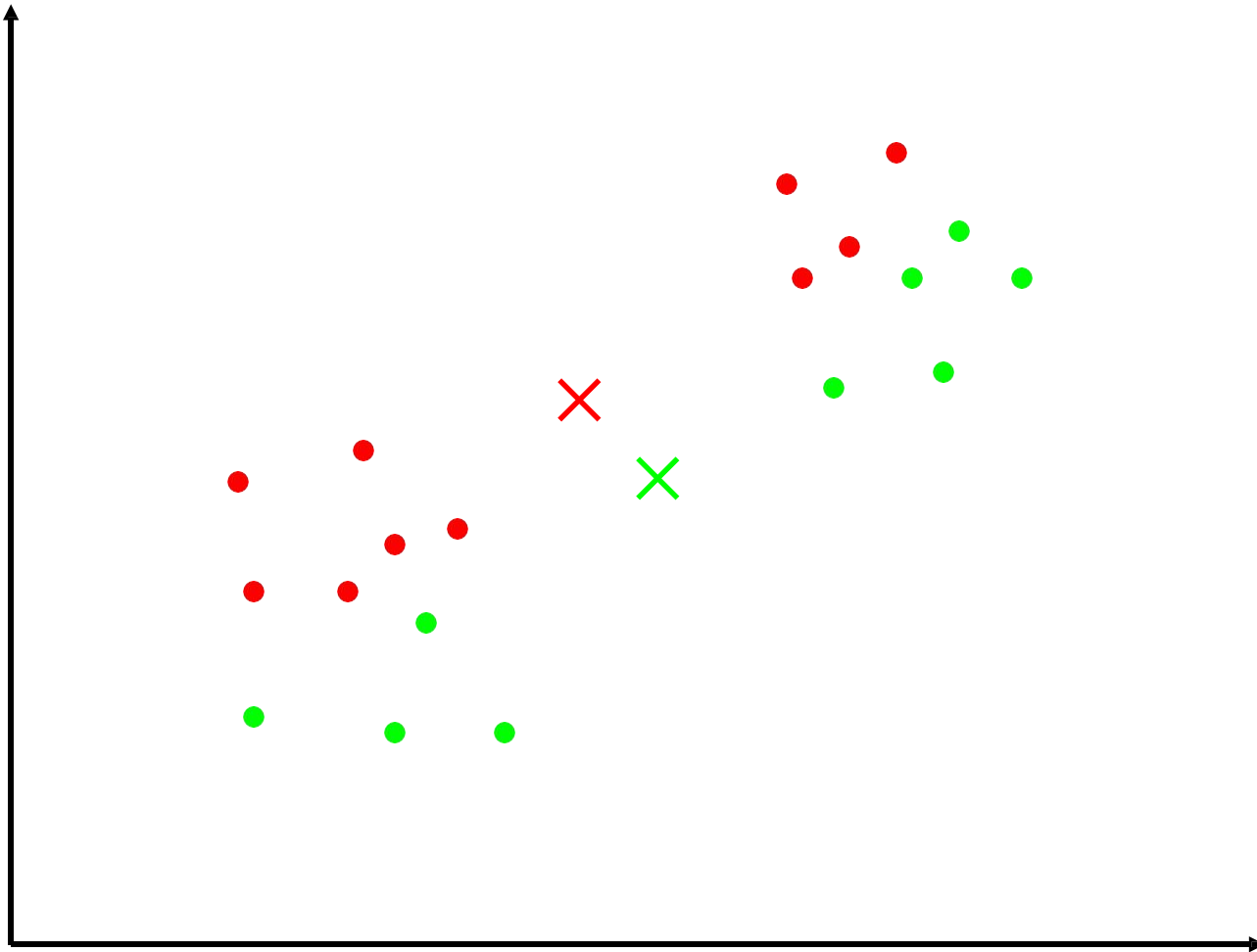
K-Means Clustering



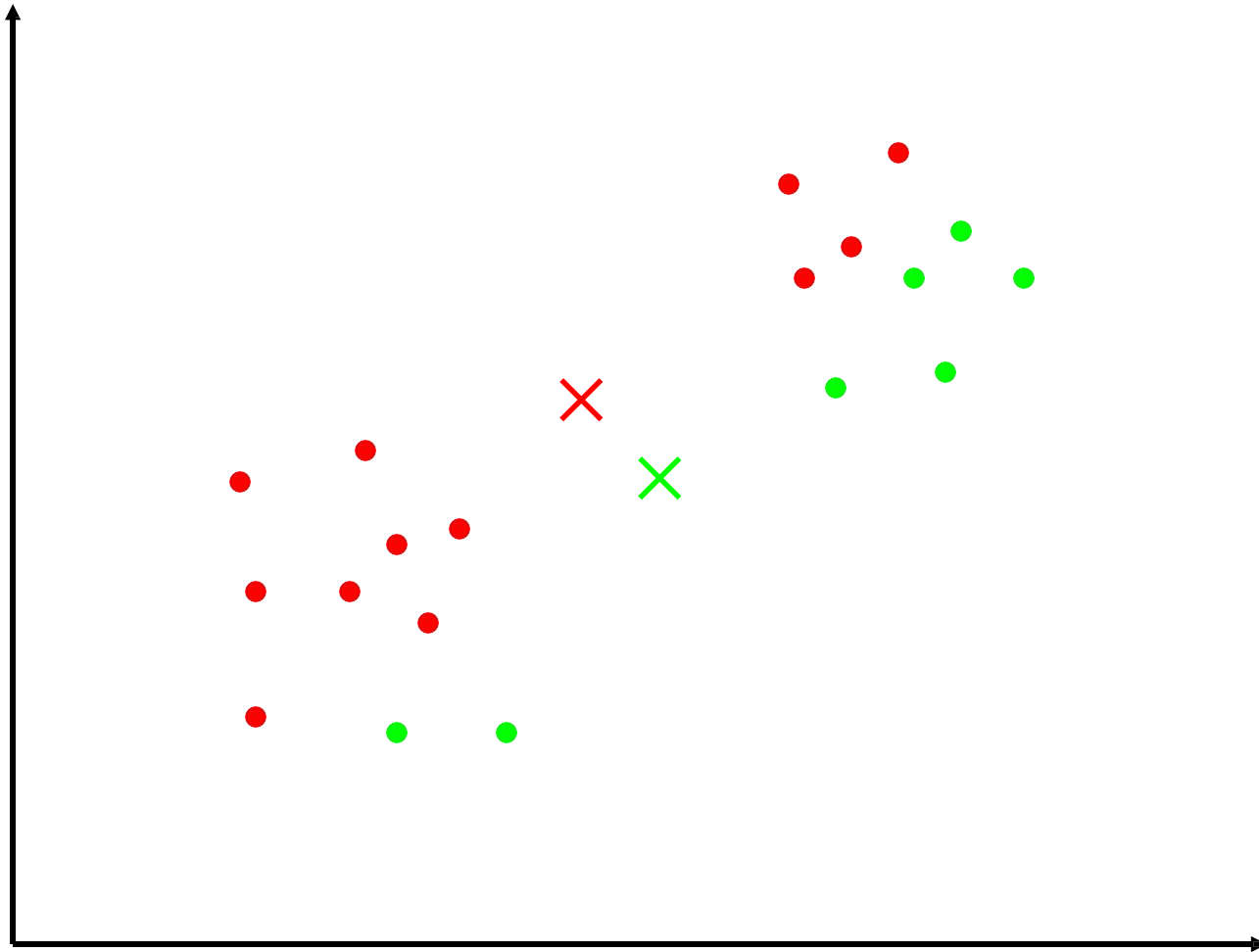
K-Means Clustering



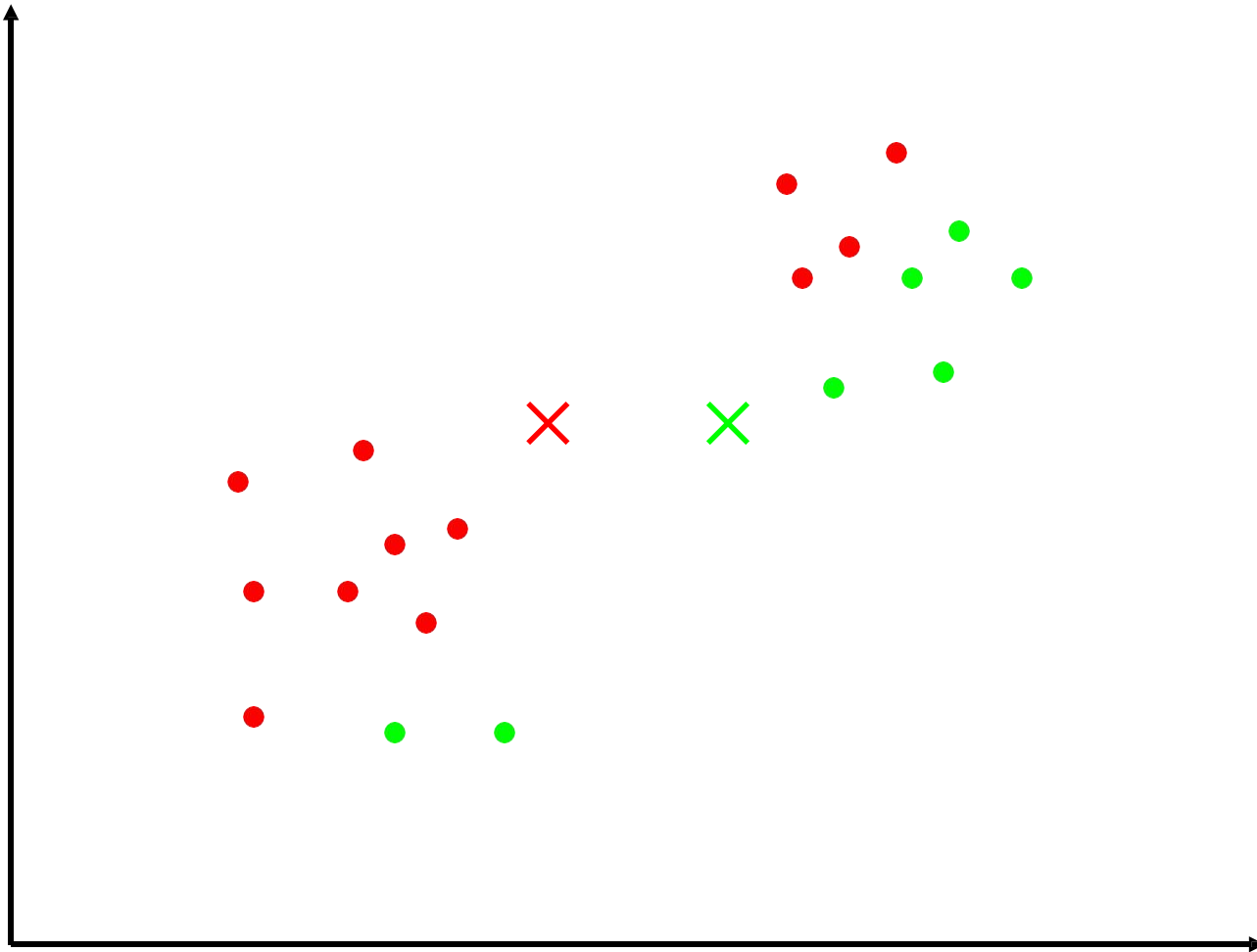
K-Means Clustering



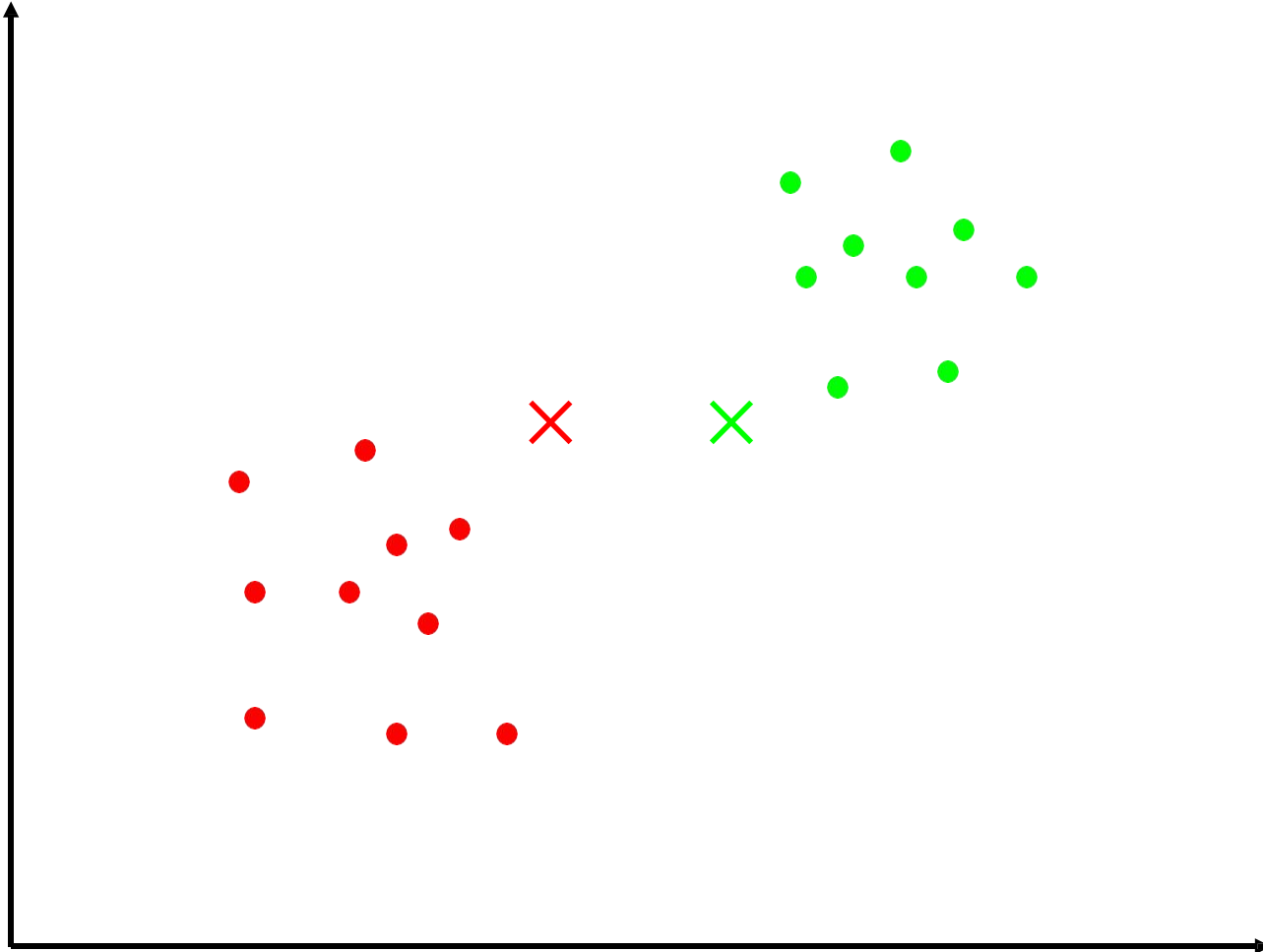
K-Means Clustering



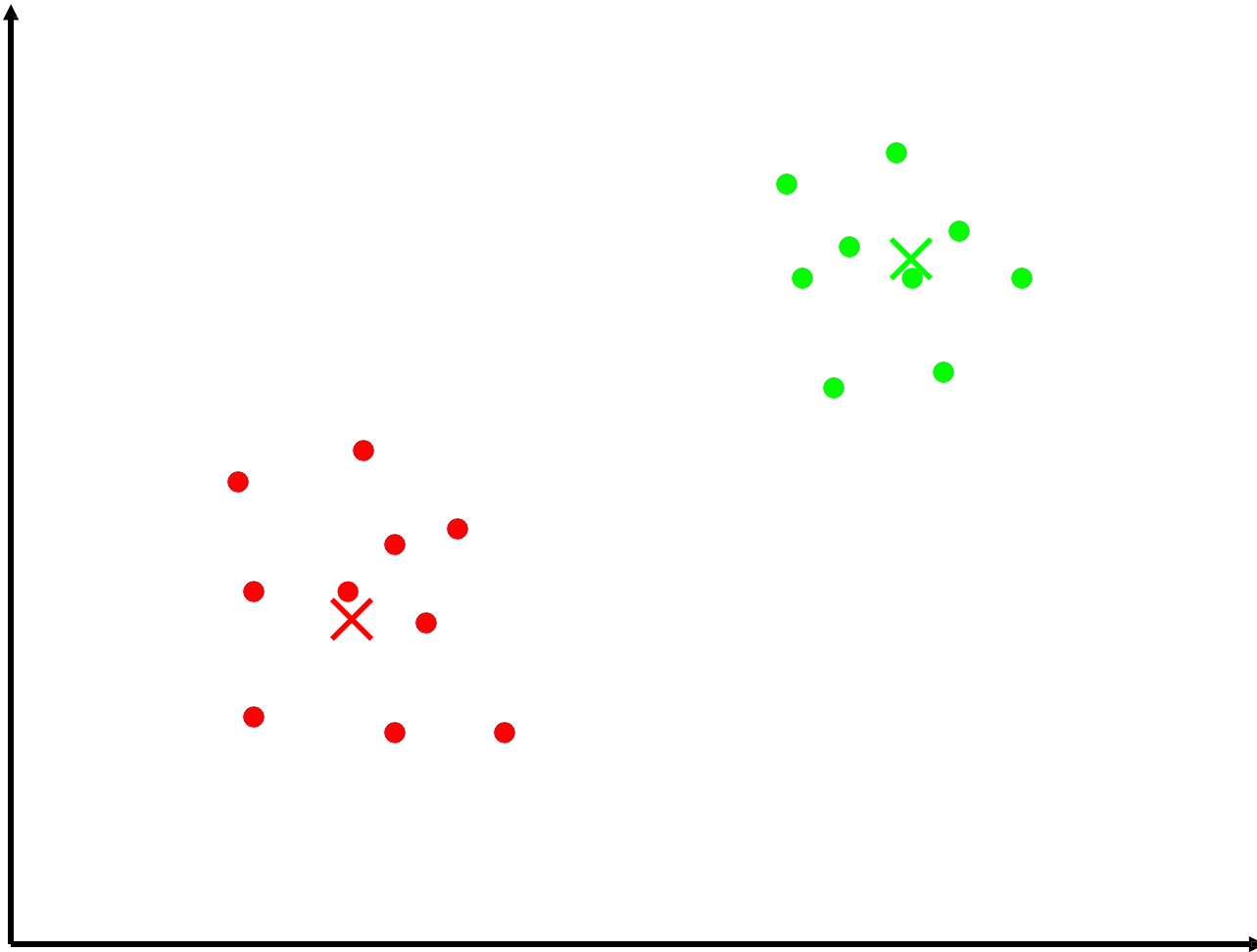
K-Means Clustering



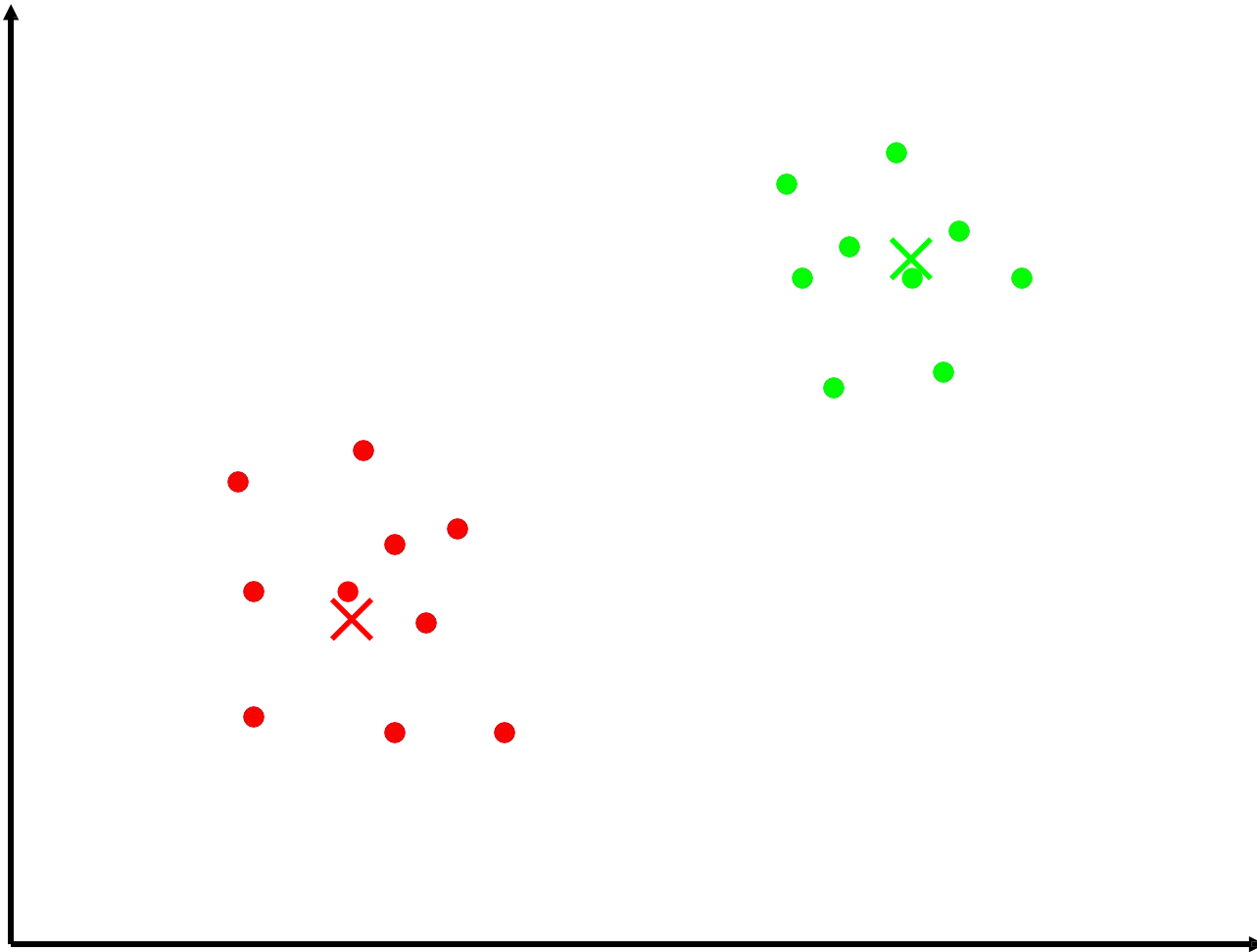
K-Means Clustering



K-Means Clustering

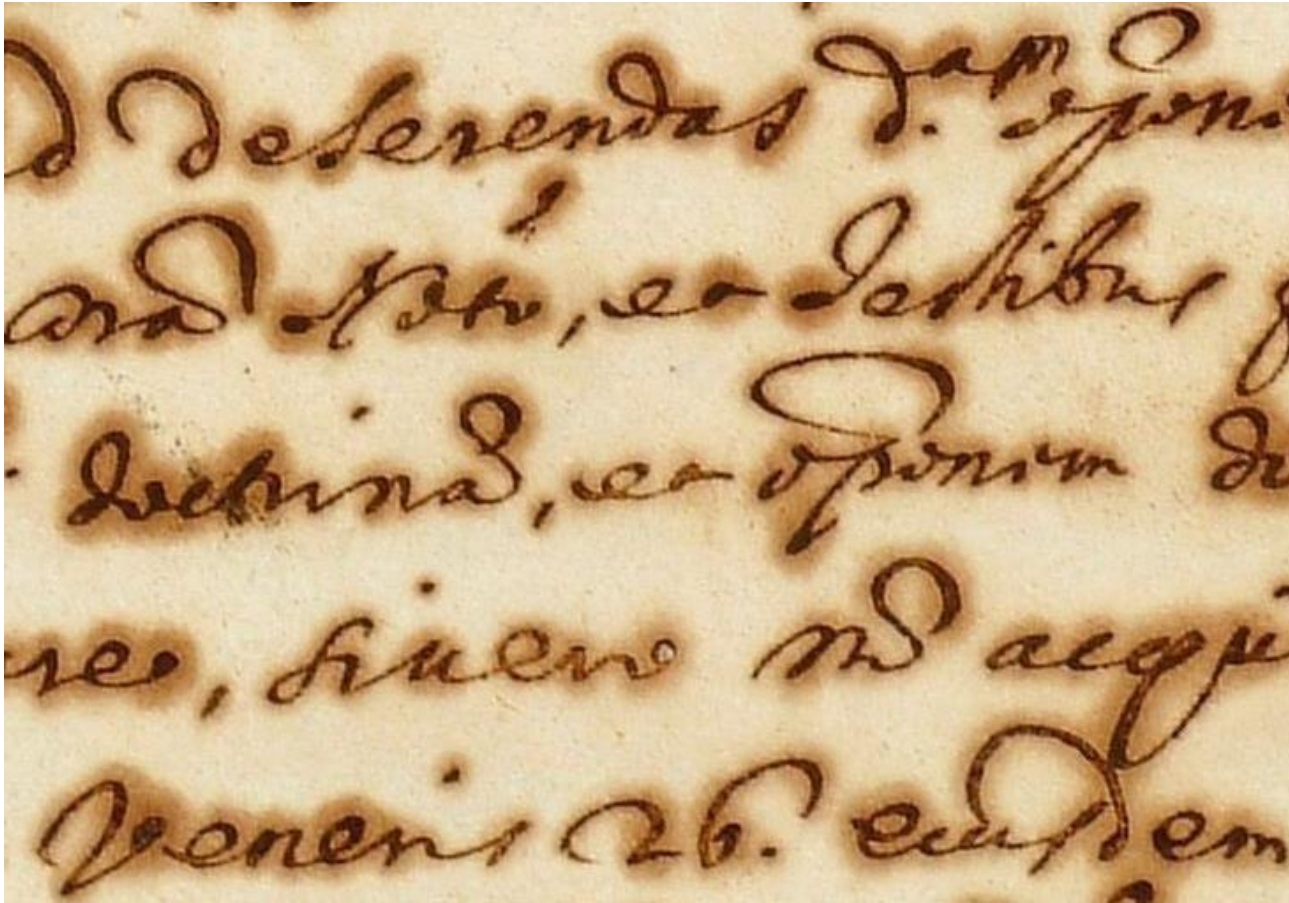


K-Means Clustering



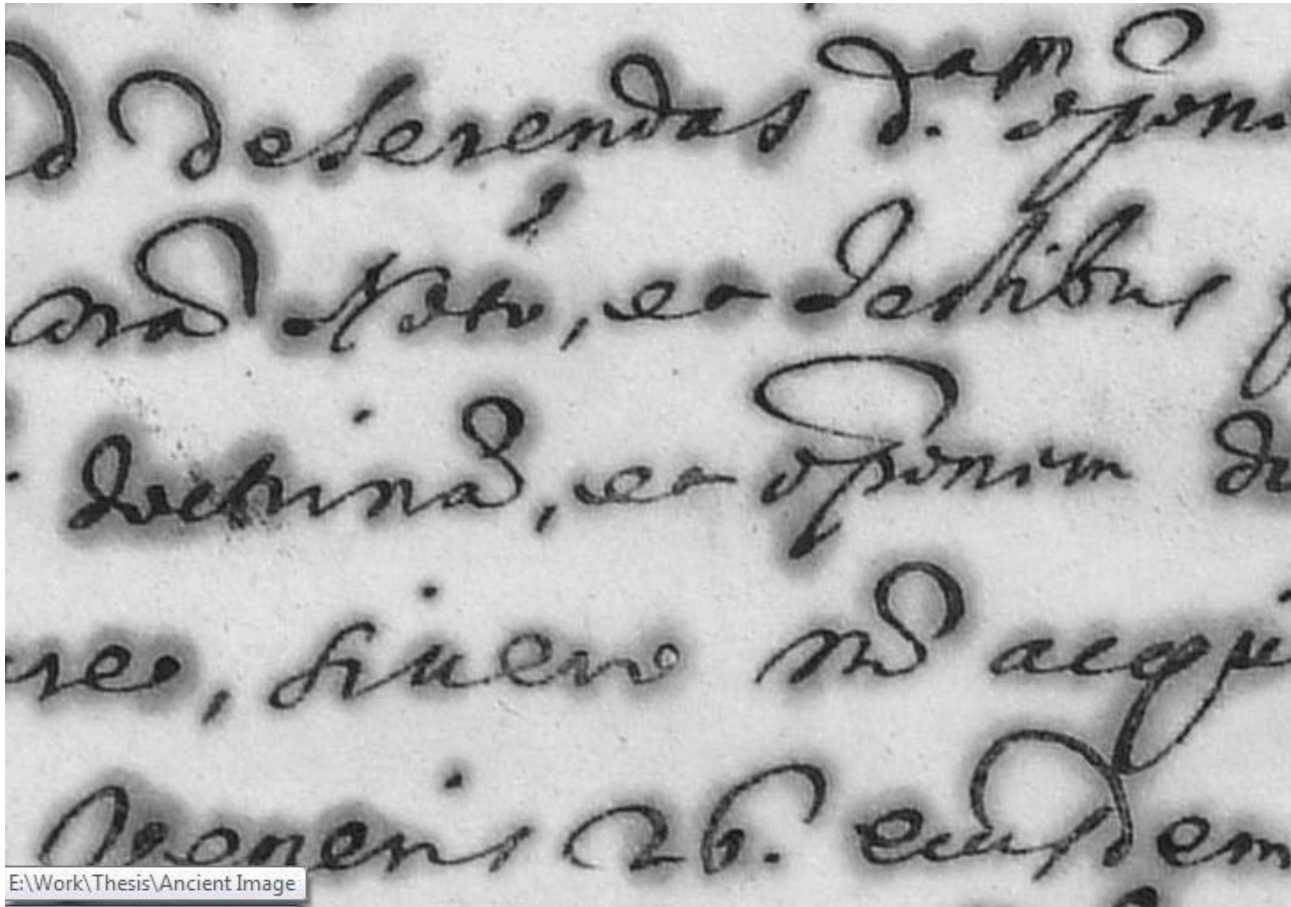
Clustering

- Example



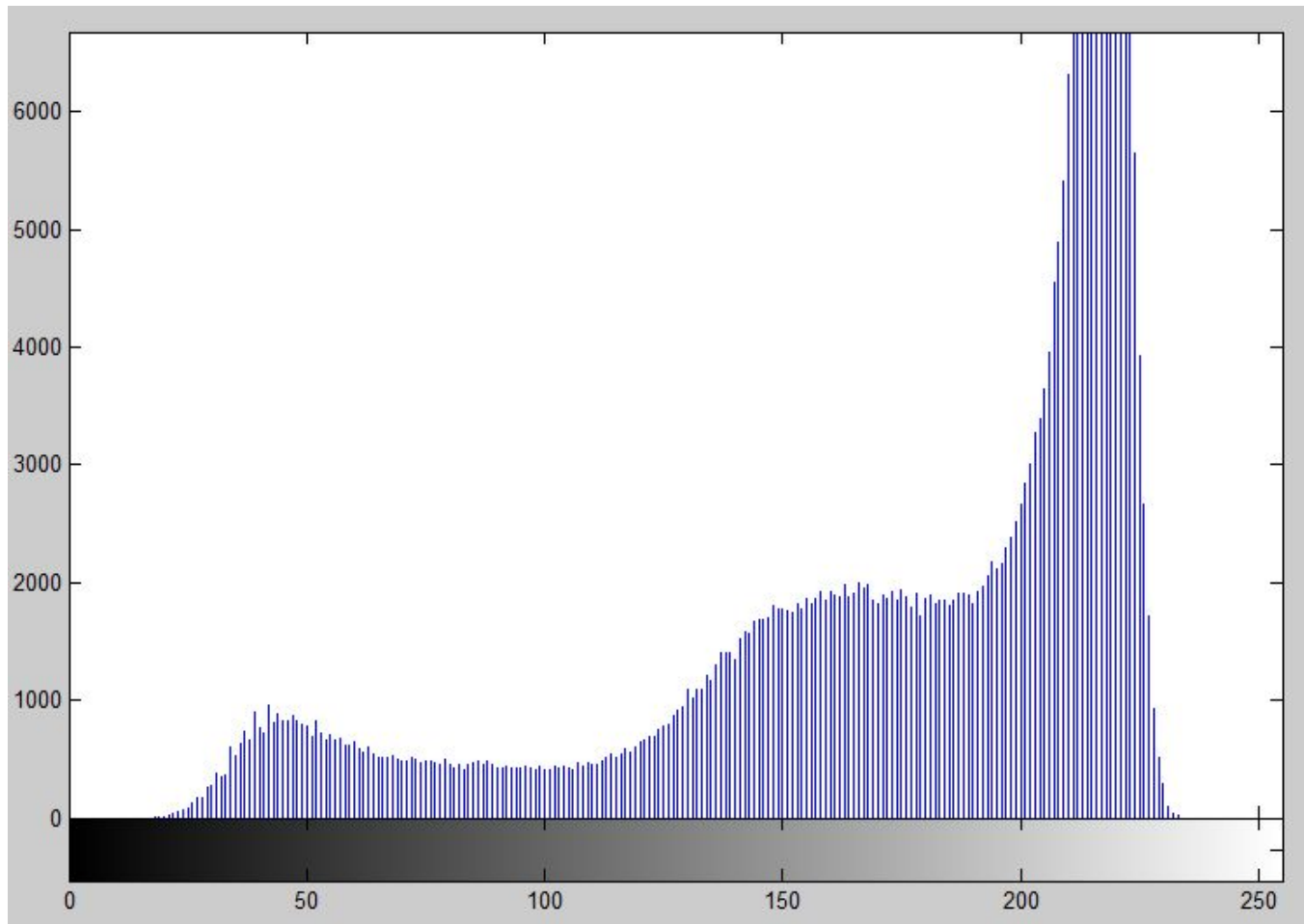
Clustering

- Example



Clustering

- Example



Clustering

- Example



Clustering

- Example



D. Comaniciu and P. Meer, *Robust Analysis of Feature Spaces: Color Image Segmentation*, 1997.

K-Means Clustering

- Example



Original



K=5



K=1
1

Acknowledgements

- ◆ Digital Image Processing”, Rafael C. Gonzalez & Richard E. Woods, Addison-Wesley, 2002
- ◆ Computer Vision: Algorithms and Applications Richard Szeliski
- ◆ Digital image processing by Dr Usman Akram